

Automatic Gas Regulator with Gas Leakage Detection & Auto -Shut-Off System

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Abstract : PROBLEM STATEMENT

LPG cylinders are widely used in households and commercial kitchens. Gas leakage due to faulty regulators, damaged pipes, or improper handling can lead to serious hazards such as fires and explosions. Leakage is often detected only after gas concentration becomes high, increasing safety risks.



OBJECTIVES

- Real-time detection of LPG leakage using gas sensing technology
- Automated shut-off of gas supply using a servo-driven crank-slider mechanism
- Integrated safety system with alert indication and manual override control

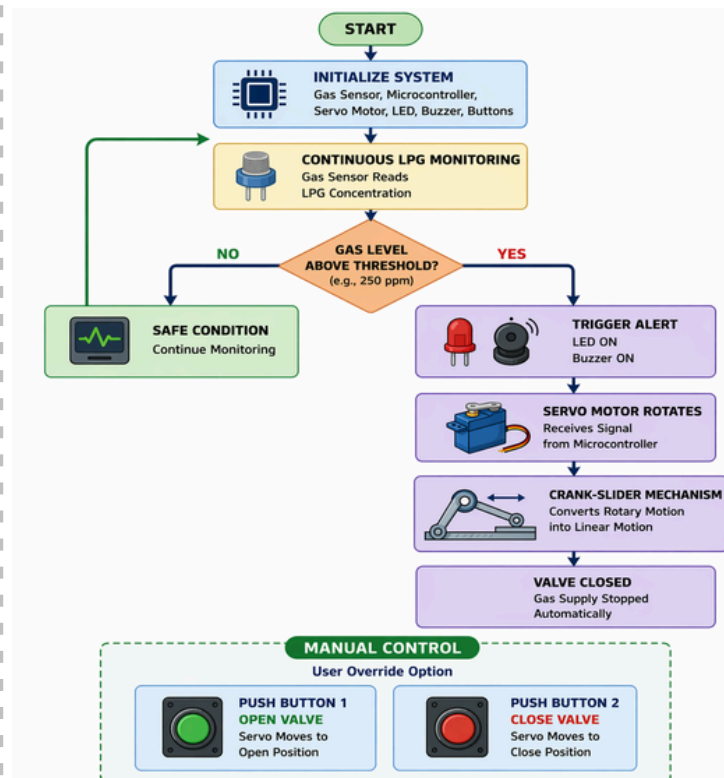
ASSUMPTIONS

- LPG concentration above 250 ppm is considered hazardous for detection and response
- Servo motor capacity is sufficient to actuate the valve via the crank-slider mechanism.

CONCEPT DESIGN

The proposed system consists of two main sections –

- a) Gas Leakage Detection
- b) Auto-Shut off Mechanism

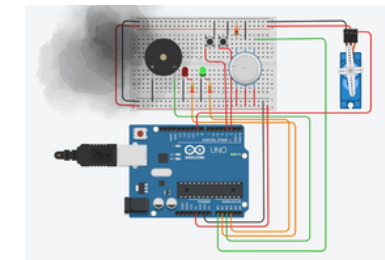


The crank-slider mechanism converts rotary motion → linear motion, where the rotation of the crank drives the connecting rod, resulting in reciprocating movement of the slider. This motion is reversible, allowing linear input to produce rotary motion. The mechanism can operate in either in-line or offset configurations depending on design requirements.

BILL OF MATERIALS

S.No	Components - Specification	Quantity
1	Arduino UNO	1
2	SG90 Micro Servo Motor	1
3	MQ-G Gas Sensor	1
4	Electromagnetic Buzzer 2v	1
5	Breadboard and Jumper Wires , Push Buttons	
6	3D Model (T-Shape Valve , Required Attachments)	

CIRCUIT DIAGRAM



3D MODEL

